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Small Animal Internal Medicine Cardiology

Prognostic Value of Intrarenal Venous Flow Analysis Using Pulsed-Wave Doppler

Tomoya Morita | Hiroki Terukina | Masahiro Yamasaki

Laboratory of Veterinary Small Animal Internal Medicine, Cooperative Department of Veterinary Medicine, Faculty of Veterinary Medicine, Iwate University, Morioka, Japan

Correspondence: Tomoya Morita (tmorita@iwate-u.ac.jp)

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ABSTRACT

Background: The predictive value of intrarenal Doppler ultrasonography (IRD) analysis using pulsed-wave Doppler for survival in dogs with myxomatous mitral valve disease (MMVD) remains unknown.

Hypothesis/Objectives: Impaired IRD indices, such as a high venous impedance index (VII), are associated with a short survival time in dogs with MMVD.

Animals: Seventy-one client-owned dogs with MMVD.

Methods: Prospective cohort study. Dogs diagnosed with various stages of MMVD between July 2019 and February 2024 were enrolled. Echocardiography and an IRD analysis were performed. Resistance index and VII were estimated using the following formula: (maximum flow velocity—minimum flow velocity)/maximum flow velocity from renal interlobar arteries and veins, respectively. Intrarenal venous flow waveforms were categorized into continuous and discontinuous patterns. Discontinuous IRVF pattern is defined as a minimum velocity of 0. The primary endpoint was defined as a composite of cardiac- and renal-related death.

Results: During the follow-up period, 18 dogs met the primary endpoint. A multivariable Cox proportional hazard analysis adjusted for the Mitral INsufficiency Echocardiographic score and plasma creatinine concentrations identified VII > 0.37 (hazard ratio, at least 4.62; 95% CI, 1.47–14.56) and discontinuous intrarenal venous pattern (hazard ratio, at least 3.62; 95% CI, 1.22–10.73) as an independent predictors of cardiac- and renal-related death. No association between resistance index and outcomes was observed in the multivariable analysis.

Conclusions and Clinical Importance: Worse outcomes in dogs with MMVD could be predicted by an intrarenal venous flow analysis.

Abbreviations: ACVIM, American College of Veterinary Internal Medicine; FAC, fractional area change; IRD, intrarenal Doppler ultrasonography; IRVF, intrarenal venous flow; LV, left ventricular; MINE, Mitral Insufficiency Echocardiographic; MMVD, myxomatous mitral valve disease; nRAA, normalized right atrial area; PH, pulmonary hypertension; RA, right atrial; RI, resistance index; RV, right ventricular; SDMA, symmetric dimethylarginine; TR, tricuspid regurgitation; VII, venous impedance index.

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1 | Introduction

Close and complementary interactions between heart failure and renal impairment, known as cardiorenal syndrome, are recognized in human medicine. In dogs with myxomatous mitral valve disease (MMVD), the prevalence of renal disease increases with the severity of heart disease [1–3]. Furthermore, survival times are significantly shorter in dogs with MMVD and renal disease than in those with MMVD alone [3]. Renal congestion caused by an elevated right atrial (RA) pressure and decreased right ventricular (RV) function is a major factor contributing to cardiorenal syndrome in human patients with heart failure [4–6]. Because the kidney is surrounded by a rigid capsule, interstitial congestion of the kidney compresses intrarenal structures (i.e., veins, glomeruli, and tubules), diminishing renal function (the renal tamponade hypothesis) [7]. The assessment of renal congestion is important for the management of heart failure.

Renal congestion reduces vessel compliance due to compression of intrarenal vessels in the renal parenchymal regions, resulting in changes in intrarenal blood flow. Intrarenal Doppler ultrasonography (IRD) using pulsed-wave Doppler has been reported to be a diagnostic modality associated with recognizing renal congestion [8] and is a promising tool for assessing prognosis in human heart failure patients [9–12]. Intrarenal venous flow (IRVF) patterns are considered as useful predictors of outcomes. A discontinuous IRVF pattern is strongly associated with poor clinical outcomes in human patients with heart failure [9–12].

The resistance index (RI) based on renal artery flow has been the focus of IRD in dogs [13, 14], and studies on renal vein flow have yet to be conducted. In dogs without heart disease, a recent study established the reference interval of the venous impedance index (VII) based on renal vein flow and showed that the IRVF pattern was continuous in all study dogs [15]. However, VII and IRVF patterns have not been studied in dogs with MMVD. Therefore, the present study aimed to clarify the determinants associated with IRD indices using pulsed-wave Doppler and investigate the prognostic value of IRD indices in dogs with MMVD. We hypothesized that IRD indices, particularly VII and IRVF patterns, change with the severity of disease and are associated with a worse prognosis in dogs with MMVD as per a composite endpoint of reaching either cardiac or renal death (i.e., cardiorenal death).

2 | Materials and Methods

This was a prospective, single-center, cohort study that included dogs with MMVD and was approved by the Animal Research Committee of Iwate University (Approval No. A201941). Informed consent was obtained from all dog owners involved in this study.

2.1 | Animals

Client-owned MMVD dogs were prospectively enrolled in the present study between July 2019 and February 2024 at the Iwate University Veterinary Teaching Hospital. Dogs were diagnosed with MMVD if they had a left apical systolic murmur and echocardiographic evidence of mitral valve leaflet thickening and/or mitral valve prolapse in the B-mode accompanied by any degree of mitral regurgitation on color Doppler imaging. Dogs of any age were included. Dogs with a body weight > 15 kg, clinically relevant systemic diseases such as any neoplasia, International Renal Interest Society stage 3 and 4 chronic kidney disease (a serum creatinine concentration > 2.8 mg/dL), adrenal or thyroid dysfunction, atrial fibrillation or ventricular tachycardia, or corticosteroid therapy were excluded. Dogs with MMVD underwent thoracic radiography and echocardiography and were classified into stages B1, B2, or C based on the American College of Veterinary Internal Medicine (ACVIM) consensus statement guidelines [16]. Stage C included MMVD dogs with current or past clinical and radiographic evidence of cardiogenic pulmonary edema or right-sided congestive heart failure. Right-sided congestive heart failure was diagnosed if the presence of ascites was associated with a subjectively dilated caudal vena cava without or with minimal inspiratory collapse. The probability of pulmonary hypertension (PH) was assessed using echocardiography based on the ACVIM consensus statement guidelines [17]. Dogs showing a high probability of PH were diagnosed with PH in the present study.

2.2 | Biochemical Analysis and Urinalysis

Feed was withheld from all dogs for 12 h before blood sampling. Blood urea nitrogen and creatinine concentrations were measured in plasma after centrifugation using an automated biochemical analyzer (Catalyst One; IDEXX Laboratories Inc., Westbrook, ME). Plasma symmetric dimethylarginine (SDMA) concentration was determined by the reference laboratory (IDEXX Laboratories, Tokyo, Japan). These analyses were performed at the same time as echocardiography.

2.3 | Echocardiographic Measurements

Echocardiography was performed by one experienced investigator (T.M.) using a commercially available ultrasound unit (Aplio a Verifia; Canon Medical Systems Corporation, Tochigi, Japan) with a sector phased array transducer frequency of 2.8–6.2 MHz (PST-50BT; Canon Medical Systems Corporation, Tochigi, Japan). All dogs were positioned in left and right lateral recumbency without sedation. A simultaneous ECG trace (lead II) was recorded.

Left ventricular (LV) dimensions were measured from a right parasternal short-axis view at the level of the chordae tendineae using the M-mode imaging. The normalized dimension was calculated as per the following equation [18]: LV end-diastolic diameter normalized for body weight = LV end-diastolic diameter/(body weight)$^{0.294}$. Fractional shortening was calculated as per the following equation: Fractional shortening = (LV end-diastolic diameter—LV end-systolic diameter)/LV end-diastolic diameter × 100%. The diameters of the left atrium and aorta were obtained from a right parasternal short-axis view at the level of the left atrium at end-systole, and the left atrium to aorta root diameter ratio was calculated. The transmitral early diastolic wave velocity was recorded on pulsed-wave Doppler with a left apical four-chamber view. The Mitral INsufficiency

Echocardiographic (MINE) score was calculated based on the values obtained for LV end-diastolic diameter normalized for body weight, fractional shortening, left atrium to aorta root diameter ratio, and transmitral early diastolic wave velocity by Vezzosi et al. [19]. The peak tricuspid regurgitation velocity was measured from multiple views, one of which provided the highest velocity that was eventually recorded on continuous-wave Doppler. From the RV-focused apical four-chamber view, which included the RV apex, RV end-diastolic area and end-systolic area were measured by manually tracing the RV endocardium at end-systole and end-diastole, respectively, from the annulus to the apex, and the fractional area change (FAC) was calculated as (RV end-diastolic—RV end-systolic area)/RV end-diastolic area × 100% [20]. The RV Tei index was calculated as the sum of the two isovolumetric periods divided by the ejection time using tissue Doppler [21]. The RA area was measured by manually tracing from the lateral aspect of the tricuspid annulus to the septal aspect from a left apical four-chamber view at end-systole, and the normalized RA area (nRAA) was calculated according to the following equation: nRAA = RA area/body surface area [22].

2.4 | Intrarenal Doppler Ultrasonographic Measurements

Intrarenal blood flow in the left kidney was recorded using pulsed-wave Doppler in right lateral recumbency using the same phased array, 2.8–6.2 MHz transducer used for echocardiography. The investigator was not blinded to the dog's echocardiographic data. Interlobar arteries and veins were detected using color Doppler imaging. The velocity range of color Doppler was set to 16.0–19.2 cm/s, and the sample volume was set at the interlobar vessels. Interlobar artery and vein waveforms were recorded simultaneously (Figure 1). Suitable interlobar vessels were selected to set the sample volume parallel to the interlobar vessels. The RI and VII were calculated by measuring maximum and minimum flow velocities as follows [11]: RI = (maximum flow velocity—minimum flow velocity)/maximum flow velocity from arteries, VII = (maximum flow velocity—minimum flow velocity)/maximum flow velocity from veins. The average of three nonconsecutive three measurements during sinus rhythm was used. Intrarenal venous waveforms were categorized into continuous and discontinuous IRVF patterns. A discontinuous IRVF pattern is defined as a minimum velocity of 0 [11]. In a previous repeatability and reproducibility study using dogs without heart disease, both the within-day and interobserver intraclass correlation coefficients were high (RI, 0.90 and 0.92; VII, 0.96 and 0.79) [15].

2.5 | Outcome Assessment

All dogs were followed up from the day of the first echocardiogram at their inclusion. An investigator (T.M.) reviewed medical records and conducted telephone interviews with dog owners to assess outcomes and identify the cause of death. The primary endpoint of the present study was a composite of cardiac- and renal-related death (cardiorenal death). Cardiac-related death was defined as spontaneous death secondary to

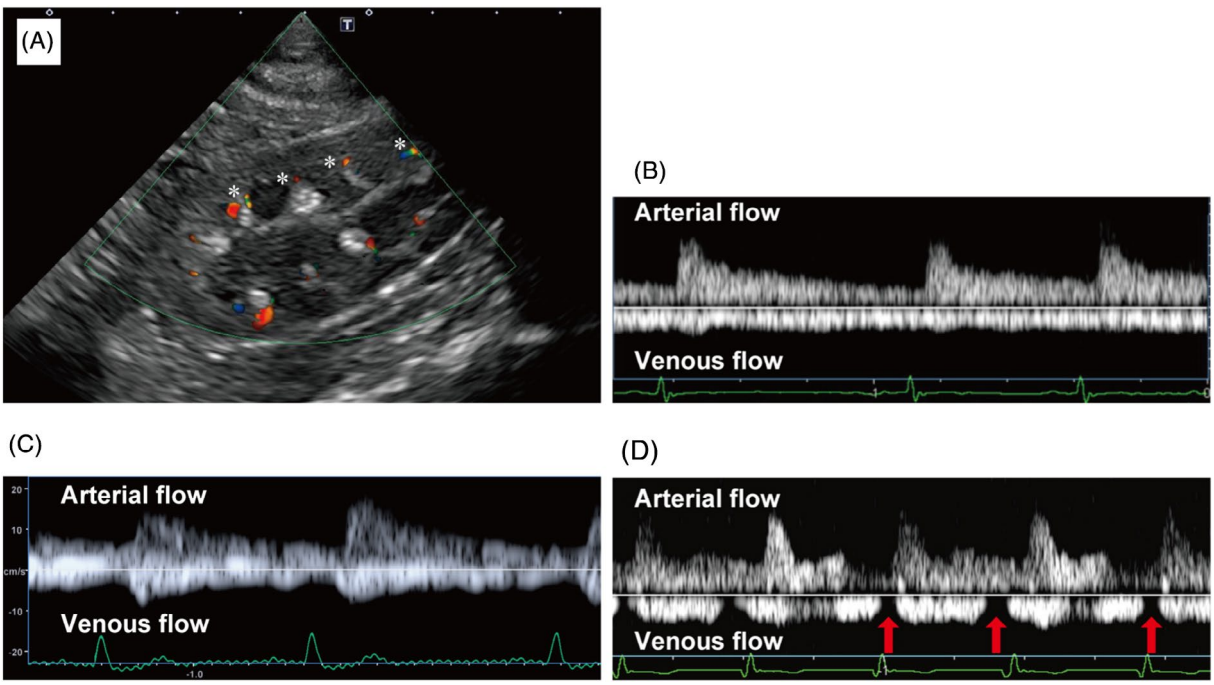

FIGURE 1 | Images illustrating the technique used to perform intrarenal Doppler ultrasonography (IRD) and representative image of dogs with MMVD. (A) Color Doppler flow image from the left kidney. The sample volume was set in the interlobar vessels (*). (B) Pulsed-wave Doppler waveform from the interlobar vessels. The upward Doppler signal shows interlobar arterial flow and the downward Doppler signal shows interlobar venous flow. Intrarenal venous flow shows a continuous pattern. (C) A representative image of IRD from a 12-year-old male Maltese with stage C and a high probability of pulmonary hypertension. Intrarenal venous flow shows a continuous pattern and VII > 0.37. The value of VII was 0.61. (D) A representative image of IRD from a 9-year-old castrated male Cavalier King Charles Spaniel with stage D and a high probability of pulmonary hypertension. Intrarenal venous flow was disrupted (red arrows) and subdivided into a discontinuous pattern (VII = 1.0).

cardiogenic pulmonary edema or euthanasia because of refractory heart failure or sudden death when no other cause was evident. Spontaneous death due to refractory right-sided congestive heart failure (ascites or pleural effusion) was also included as cardiac-related death. Renal-related death was defined as spontaneous death or euthanasia secondary to progressive azotemia (progression to International Renal Interest Society stage 3 or 4 on blood examination at least every 3 months) without abnormalities in the respiratory status or right-sided congestive heart failure. Death from any other cause was defined as noncardiorenal death. Survival times were calculated as the interval from the day of the first echocardiogram at inclusion to either the day of death or the closing time of the study (February 2024). Dogs were censored in the survival analysis if they were alive at the end of the study period, died due to noncardiorenal death, or underwent mitral valve repair. Dogs that were lost to follow-up were right-censored after the last contact.

2.6 | Statistical Analysis

Statistical analyses were performed using JMP version 14.0 (SAS Institute Inc., Cary, NC). The normal distribution of data was confirmed by the Shapiro–Wilk test. Continuous data are reported as the mean ± standard deviation for normally distributed data and as the median (interquartile range) for nonnormally distributed data. Two-group comparisons were performed with unpaired *t* tests for normally distributed data and with the Wilcoxon rank-sum test for nonnormally distributed data. The Fisher's exact test was used to examine the significance of differences in categorical variables. Multiple comparisons between more than two groups were performed with the Tukey–Kramer test for normally distributed data and with the Steel-Dwass test for nonnormally distributed data. Dogs were dichotomized based on previously reported cut-off values for RI or VII [15]. The upper 95% limit was used as the cut-off value: RI > 0.74 and VII > 0.37. To determine the relationship between RI or VII and the characteristics and echocardiographic variables, Pearson's correlation coefficient was used. Correlation was reported as indicating poor to very strong correlation: < 0.2 is considered as poor, 0.21–0.60 as fair, 0.61–0.80 moderate, and > 0.8 as very strong [23].

Survival analyses were performed using a Kaplan–Meier analysis and compared using the log-rank test. If half of the dogs or more did not reach the primary endpoint during the follow-up period, median survival time was defined as not determinable (ND). The independent prognostic power of IRD variables (RI > 0.74, VII > 0.37, and a discontinuous IRVF pattern) was assessed using univariable and multivariable Cox proportional hazard analyses. Covariates in the multivariable model were selected from previously reported predictors of a poor outcome in dogs with MMVD [3, 19, 24–25]. The variance inflation factor was used to detect multicollinearity among covariates. Variables with the variance inflation factor exceeding a predefined threshold (variance inflation factor = 5) were indicative of multicollinearity. To avoid overfitting, the number of covariates incorporated into the multivariable model was limited to three, and two independent multivariable Cox hazard models were constructed: (i) model 1 adjusted for the MINE score and the RV Tei index; (ii) model 2 adjusted for the MINE score and plasma creatinine concentrations. The Cox proportional hazard model assumption was checked via the Schoenfeld residual.

The threshold for significance was set at a $p < 0.05$.

3 | Results

3.1 | Characteristics

The present study included 71 dogs from 15 breeds, with Chihuahua being the most frequently represented ($n = 16$), followed by Toy Poodle ($n = 13$), mixed breeds ($n = 10$), Cavalier King Charles Spaniel ($n = 9$), Pomeranian ($n = 6$), Shih Tzu ($n = 3$), Chin ($n = 3$), Maltese ($n = 2$), Miniature Schnauzer ($n = 2$), Papillon ($n = 2$), Miniature Dachshund ($n = 1$), Beagle ($n = 1$), Boston Terrier ($n = 1$), Yorkshire Terrier ($n = 1$), and Shiba Inu ($n = 1$). Mean body weight and age were 5.3 ± 2.7 kg and 11.4 ± 2.4 years, respectively. Dogs were classified into stage B1 ($n = 26$), stage B2 ($n = 28$), and stage C ($n = 17$). The dogs in stage C included left-sided and right-sided congestive heart failure ($n = 4$), left-sided congestive heart failure ($n = 9$), and right-sided congestive heart failure ($n = 4$). The echocardiographic variables in different ACVIM stages are shown in Table S1. Twenty-two dogs (22/71) were diagnosed with PH (one stage B1; eight stage B2; 13 stage C) and eight (8/71) had ascites due to right-sided congestive heart failure. The mean TR velocity was 4.0 ± 0.8 m/s in dogs diagnosed with PH. Twenty-nine dogs were prescribed angiotensin-converting enzyme inhibitors (seven stage B1; 12 stage B2; 10 stage C), 41 pimobendan (four stage B1; 20 stage B2; 17 stage C), and 21 diuretics (two stage B1; four stage B2; 15 stage C). Diuretics included furosemide (18/21), torsemide (3/21), and spironolactone (6/21). Three dogs received sildenafil citrate. In this study, two stage B1 dogs were prescribed on diuretics by the referring hospital at the time of the initial examination, and two stage C dogs were not prescribed on diuretics because their first visit at our hospital was the date of inclusion. Mean plasma blood urea nitrogen and creatinine concentrations at inclusion were 25.3 ± 11.8 mg/dL (reference range, 7–27 mg/dL) and 0.95 ± 0.36 mg/dL (reference range, 0.5–1.8 mg/dL), respectively.

3.2 | IRD Findings

The mean RI value was 0.69 ± 0.06 (reference interval, 0.55–0.74) and the median VII value was 0.27 (0.22–0.33; reference interval, 0.13–0.37). The RI value was significantly higher in ACVIM stage B2 (0.70 ± 0.06, $p = 0.02$) and C (0.72 ± 0.07, $p < 0.001$) dogs than in stage B1 dogs (0.66 ± 0.06; Figure 2). The VII value was significantly higher in ACVIM stage C dogs (0.53 [0.31–1.00]) than in stage B1 (0.24 [0.21–0.29], $p < 0.001$) and B2 dogs (0.27 [0.21–0.28], $p < 0.001$; Figure 2). Thirteen dogs had RI > 0.74. Thirteen dogs had VII > 0.37 and seven had VII > 0.37 without a discontinuous IRVF pattern. Five dogs had both a high value of RI and VII. Four dogs with RI > 0.74 showed a discontinuous IRVF pattern. Dogs were subdivided into those showing continuous ($n = 65$) and discontinuous ($n = 6$) IRVF patterns. All dogs showing a discontinuous IRVF pattern belonged to stage C and presented with right-sided

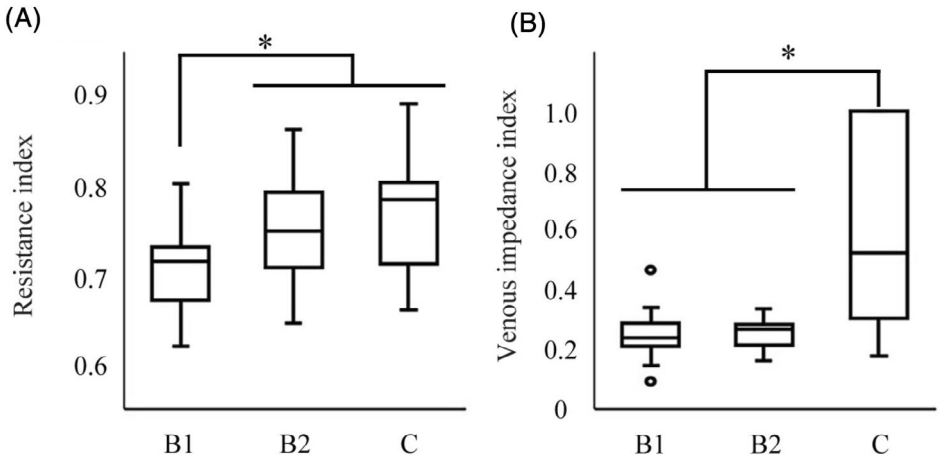

FIGURE 2 | Box-and-whisker plots of RI and VII in dogs with myxomatous mitral valve disease. (A) Resistance index was significantly higher in ACVIM stage B2 and C dogs than in stage B1, (B) Venous impedance index was significantly higher in ACVIM stag C dogs than in stage B1 and B2. *$p < 0.05$.

congestive heart failure. Eight dogs with ascites were categorized as showing a discontinuous (6/8) and a continuous (2/8) IRVF pattern, respectively. In comparison with dogs showing a continuous IRVF pattern, those showing a discontinuous IRVF pattern had high blood urea nitrogen and SDMA concentration, a high MINE score, and a high TR velocity, large RA size, low RV-FAC, and high RV Tei index (Table S2).

3.3 | Correlation Analysis

There was a fair correlation between RI and TR velocity ($r = 0.47$, $p < 0.001$), RV-FAC ($r = -0.38$, $p < 0.001$), and nRAA ($r = 0.49$, $p < 0.001$). There was a fair correlation between VII and MINE score ($r = 0.33$, $p = 0.01$), RV-FAC ($r = -0.42$, $p < 0.001$), RV Tei index ($r = 0.33$, $p = 0.01$), and SDMA ($r = 0.53$, $p < 0.001$). There was a moderate correlation between VII and TR velocity ($r = 0.70$, $p < 0.001$), and nRAA ($r = 0.77$, $p < 0.001$). Age was not correlated with RI and VII.

3.4 | Survival Analysis

During the follow-up period (median 302 days, range 5–1470 days), 18 dogs (25.4%) met the primary endpoint (14 [19.7%] died of cardiac-related causes and four [5.7%] died of renal-related causes). Dogs that met the primary endpoint included stage B1 (4/18, 22%), B2 (3/18, 17%), and C (11/18, 61%). None of the dogs were euthanatized in the present study. Nine dogs (12.7%) were considered to have died of other causes, and 44 (62.0%) were still alive at the end of the study. One dog with stage C underwent mitral valve repair on Day 38 from the time of diagnosis. This dog was censored on the day of surgery. The median survival time for all-cause death was 166 days (95% CI, 5–952 days), while that for cardiorenal death was 141 days (95% CI, 5–460 days). The median follow-up for dogs still alive was 407 days (95% CI, 41–1460 days).

The Kaplan–Meier analysis with log-rank test revealed the significant effects of RI, VII, and IRVF patterns on survival (Table 1, Figure 3). Dogs with RI > 0.74 had significantly shorter survival times ($p = 0.02$) than those with RI ≤ 0.74. Dogs with VII > 0.37 had significantly shorter survival times ($p < 0.001$) than those with VII ≤ 0.37. Dogs with a discontinuous IRVF pattern had significantly shorter survival times ($p < 0.001$) than those with a continuous pattern.

TABLE 1 | Kaplan–Meier analysis with log-rank test in 71 dogs with myxomatous mitral valve disease.

Subgroup	*n*	Median survival time (d)	95% CI	
RI				
> 0.74	13	453	117–ND	0.02
≤ 0.74	58	ND (> 1470)	ND	
VII				
> 0.37	13	117	24–292	< 0.001
≤ 0.37	58	ND (> 1470)	ND	
IRVF patterns				
Continuous	65	ND (> 1470)	ND	< 0.001
Discontinuous	6	142	7–ND	

Note: If half of the dogs or more did not reach the primary endpoint during the follow-up period, median survival time was defined as not determinable (ND). Abbreviations: CI, confidence interval; IRVF, intrarenal venous flow; ND, not determinable; RI, resistance index; VII, venous impedance index.

In the univariable Cox proportional hazard analysis, MINE score, RV Tei index, plasma creatine concentrations, RI > 0.74, VII > 0.37, and a discontinuous IRVF pattern were associated with cardiorenal death in dogs with MMVD (Table 2). The relationship between VII > 0.37 and the primary endpoint remained significant after adjustments for MINE score and the RV Tei index (model 1; $p = 0.01$); MINE score and plasma creatinine concentrations (model 2; $p = 0.01$). A discontinuous IRVF pattern was associated with the primary endpoint in model 2 ($p = 0.02$). In contrast, RI > 0.74 was not associated with the primary endpoint in either multivariable Cox model.

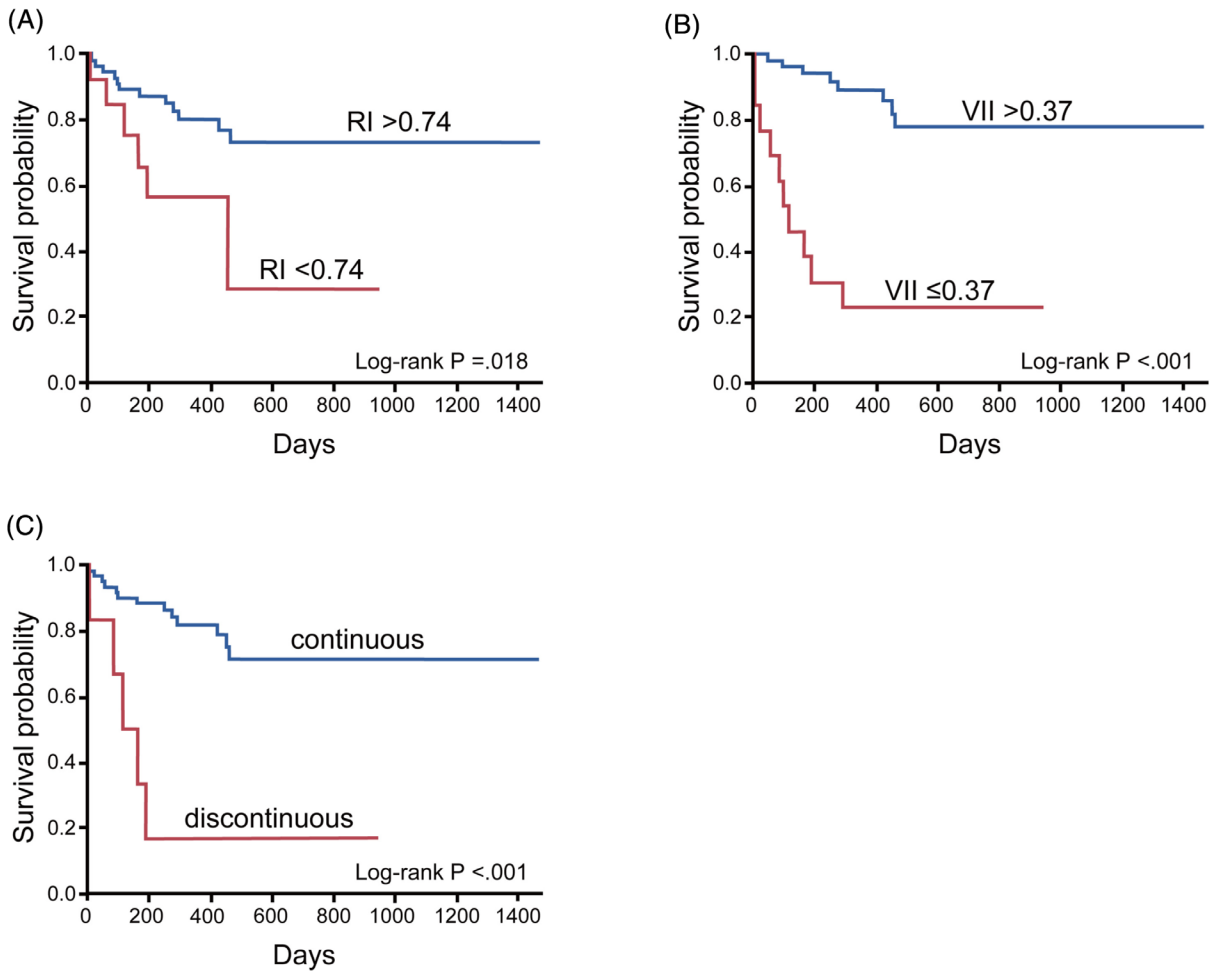

FIGURE 3 | Kaplan–Meier analysis of survival in dogs with myxomatous mitral valve disease. (A) Dogs with RI > 0.74 had worse outcomes than those with RI ≤ 0.74. (B) Dogs with VII > 0.37 had worse outcomes than those with VII ≤ 0.37. (C) Dogs with a discontinuous IRVF pattern had worse outcomes than those with a continuous pattern. IRVF, intrarenal venous flow; RI, resistance index; VII, venous impedance index.

TABLE 2 | Univariable and multivariable Cox proportional hazard analysis for the association with the cardiorenal death in 71 dogs with myxomatous mitral valve disease.

	Unadjusted		Model 1		Model 2	
Variable	**HR (95% CI)**	***p***	**HR (95% CI)**	***p***	**HR (95% CI)**	***p***
MINE score (per 1 score)	1.55 (1.25–1.96)	< 0.001				
RV Tei index (per 0.1)	1.34 (1.10–1.59)	0.01				
Serum creatinine level (per 0.1 mg/dL)	1.14 (1.01–1.28)	0.04				
RI > 0.74	3.14 (1.16–8.47)	0.02	2.01 (0.66–6.11)	0.16	2.50 (0.75–8.28)	0.14
VII > 0.37	9.89 (3.82–25.62)	< 0.001	4.36 (1.54–12.33)	0.01	4.62 (1.47–14.56)	0.01
Discontinuous IRVF patterns	6.51 (2.28–18.57)	< 0.001	3.02 (0.86–10.69)	0.09	3.62 (1.22–10.73)	0.02

Note: Multivariable Cox proportional hazard ratio analyses included three variables. Model 1: MINE score, RV Tei index, and each of IRD variables; Model 2: MINE score, serum creatinine level, and each of IRD variables.
Abbreviations: CI, confidence interval; HR, hazard ratio; IRVF, intrarenal venous flow; MINE, Mitral INsufficiency Echocardiographic; RI, resistance index; RV, right ventricular; VII, venous impedance index.

4 | Discussion

In the present study, increased VII and the presence of a discontinuous IRVF pattern were associated with a shortened survival time in dogs with MMVD. In addition, the combined assessment of VII and previously reported prognostic indices, such as the MINE score, the RV Tei index, and plasma creatinine concentrations, could further increase our ability to predict worse outcomes in dogs with MMVD. These results support our hypothesis that VII and IRVF patterns are associated with a

worse prognosis in dogs with MMVD. Since increased VII is related to a poor prognosis in dogs with MMVD, more aggressive treatment might be considered. The study reveals the prognostic value of an IRVF analysis using pulsed-wave Doppler in dogs with MMVD.

An evaluation of renal congestion by an IRVF analysis is useful in predicting outcomes and assessing the efficacy of treatment in human patients with heart failure [9–12, 26]. A discontinuous IRVF pattern is associated with elevated RA pressure and RV dysfunction in human patients with heart failure [9–11]. In the present study, elevated VII was associated with RA dilation and RV dysfunction in dogs with MMVD. The RA area is positively correlated with RA pressure in human patients with PH [27]. In addition, 75% of the dogs with right-sided congestive heart failure showed a discontinuous IRVF pattern and all dogs with right-sided congestive heart failure had VII > 0.37 in this study. This result suggests that an elevated RA pressure is associated with a discontinuous IRVF pattern and a high VII value. Although the association between RA pressure and IRVF is unknown because cardiac catheterization was not performed in the present study, elevated RA pressure or RV dysfunction might trigger a change in IRVF due to renal congestion in dogs.

A discontinuous IRVF pattern was associated with worse outcomes in human patients with heart failure [9–12, 26]. In the present study, a discontinuous IRVF pattern was associated with worse outcomes after adjusting for the MINE score and plasma creatinine concentrations in dogs with MMVD. These results suggest that a discontinuous IRVF pattern is a poor prognostic factor and worthy of evaluation in addition to previously reported poor prognostic factors (i.e., MINE score and creatinine concentrations) in dogs with MMVD, as in human patients with heart failure. However, the number of dogs with a discontinuous IRVF pattern was small in the present study ($n = 6$; 8.5%). A discontinuous IRVF pattern was more common in human patients with heart failure (43.3%–80.0%) [9, 11, 12, 26]. This difference might be attributable to the severity and chronicity of the underlying heart disease, the presence or absence of acute congestive heart failure, and differences in the extent of congestion between the left- and right-sided of the heart [9, 11, 12, 26]. Furthermore, in this study, all dogs with a discontinuous IRVF pattern presented with right-sided congestive heart failure. Although right-sided congestive heart failure rather than a discontinuous IRVF pattern might be associated with cardiorenal death, the number of dogs that met the primary endpoint was too small to reveal this point.

In human patients and rat models, VII was reduced by the treatment of heart failure and reflected renal function [26, 28]. However, the prognostic value of VII has not yet been established in human patients with heart failure because the number of patients with an abnormal VII value and a continuous IRVF pattern is very small (3.9%) [11]. In the present study, 9.9% of dogs with MMVD had increased VII (> 0.37) with a continuous IRVF pattern. Compared with human patients with heart failure, dogs with MMVD had a higher proportion of abnormal VII and a continuous IRVF pattern. Similar to the small number of dogs with a discontinuous pattern, the difference between humans and dogs might be related to the severity and chronicity of heart disease and differences in the extent of congestion between the left- and right-sided chambers of the heart. In the present study, VII > 0.37 was strongly associated with poor outcomes in the multivariable Cox proportional hazard model after adjustments for the results obtained in assessments of the left heart (MINE score), RV function (the RV Tei index), and renal function (plasma creatinine concentrations). The risk of cardiorenal death was approximately fourfold higher in dogs with VII > 0.37 than in those with VII ≤ 0.37 after adjustment by the MINE score, RV Tei index, and serum creatinine concentration. These results indicate that the addition of the VII assessment allows for a more detailed stratification of the risk of death in MMVD dogs.

As MMVD progresses, it causes left heart remodeling, an elevated LV filling pressure, and PH, resulting in a decrease in RV function [24, 25]. The MINE score, a left heart evaluation index, and the RV Tei index, an RV function index, have been identified as poor prognostic factors in dogs with MMVD [19, 24–25]. In the present study, a high VII value, indicating renal congestion, was an independent poor prognostic factor after adjusting for the MINE score and the RV Tei index. These results suggest that the prognosis of MMVD dogs is poorer when renal congestion occurs in addition to left heart and RV dysfunction.

The incidence of azotemia and an elevated SDMA concentration increased with disease severity in dogs with MMVD [2, 3, 13–14]. In comparison with dogs showing a continuous pattern, those showing a discontinuous IRVF pattern showed significantly greater blood urea nitrogen and SDMA concentrations in the present study. In addition, VII was positively correlated with SDMA concentration. This finding might indicate that renal impairment was caused not only by a compromise in cardiac output and parenchymal perfusion pressure but also by renal venous congestion in dogs with MMVD. However, whether renal impairment was actually caused by renal congestion or not is unknown. Furthermore, survival times were significantly shorter in dogs with MMVD and chronic kidney disease (serum creatinine > 1.4 mg/dL) than in those affected by MMVD alone [3]. In the present study, higher plasma creatinine concentrations were associated with cardiorenal death based on the univariable Cox proportional hazard analysis. As in a previous report [3], renal function impairment was associated with a poor prognosis. High VII and a discontinuous IRVF pattern were associated with cardiorenal death based on the multivariable Cox proportional hazard analysis adjusted for plasma creatinine concentrations in the present study. This result, as well as previous findings from human and rat model studies, suggests that VII might reflect not only renal function but also changes that occur during renal congestion, such as renal fibrosis and renal vasoconstriction due to activation of the sympathetic nervous system and the renin-angiotensin-aldosterone system [11, 28].

The prognostic value of RI was shown to be lower than that of IRVF patterns in human patients with heart failure [11]. Although RI > 0.74 correlated with shorter survival in the univariable analysis in the present study, a relationship was not noted between RI > 0.74 and a poor prognosis in the multivariable analysis. The findings of human studies showed that an

increased RI was related to an elevated RA pressure, reflecting renal congestion [11]. In the present study, RI was fairly correlated with RA size, TR velocity, and RV function. The RI might be weakly affected by RA pressure and RV function. The lack of association between RI and poor prognosis might be due to the influence of factors other than RA pressure and RV function, such as age. Indeed, a relationship between RI and age has been reported in humans [11, 29] and dogs without cardiac disease [15]. However, in the present study of dogs with MMVD, no association was observed between age and RI, suggesting that further investigation is needed.

The present study had several limitations that need to be addressed. The number of dogs reaching the primary endpoint was small. Therefore, the statistical power for assessing outcomes was low. Furthermore, the number of dogs with a discontinuous IRVF pattern was small. While a discontinuous IRVF pattern was associated with a worse outcome after adjustment for the MINE score and plasma creatinine concentrations in the present study, further validation is required to confirm the generating hypothesis due to the small number of dogs showing a discontinuous IRVF pattern. Therefore, the present results need to be interpreted with caution, and further studies are required to confirm the usefulness of IRD variables. In the present study, noncardiorenal deaths were censored. Since noncardiorenal deaths are competing events, censoring might have led to an overestimation of the cumulative incidence. In addition, the glomerular filtration rate (i.e., inulin clearance) was not measured in the present study. While plasma creatinine concentrations were used as a renal function test in this study, several factors, such as muscle mass, might have affected the results obtained. In this study, right-sided congestive heart failure was diagnosed if the presence of ascites was associated with a subjectively dilated caudal vena cava with insufficient inspiratory collapse. Since quantitative evaluation was not performed, the diagnosis of right-sided congestive heart failure might not be accurate. Moreover, dogs were administered many medications, which were not standardized during the follow-up period and might have affected IRD indices and survival times. Therefore, if simultaneous recording is not feasible, these findings might not be reproducible.

5 | Conclusions

The assessment of IRVF by pulsed-wave Doppler, such as VII and IRVF patterns, is useful for predicting a poor outcome in dogs with MMVD. Especially, VII >0.37 was a powerful predictor of cardiorenal death after adjustments for the results obtained in assessments of the left heart, RV function, and renal function.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Approved by Animal Research Committee of Iwate University (Approval No. A201941). Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Comparison of echocardiographic variables in dogs with different ACVIM stages. **Table S2:** Comparison of clinical and echocardiographic variables in dogs considering continuous and discontinuous intrarenal venous flow pattern.